DO NOW — RECALL

Thinking abstractly: models and reality

Name: _____ Class: _____ Date: _____

1. What is unit testing?

2. What is a precondition?

3. What do vertices and edges represent in a graph?

4. Why is an unweighted route graph not enough to identify the fastest journey?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Testing a small module or function independently.

2. Model answer

An input requirement that must hold before execution.

3. Model answer

Objects/locations and connections.

4. Model answer

It lacks travel-time weights.

Vocabulary — know these terms

abstraction — Retain relevant characteristics, ignore irrelevant ones.

model — Simplified representation serving a purpose.

attribute — Property stored about an entity.

assumption — Condition accepted in constructing the model.

limitation — A question the simplified model cannot answer.

NEW LEARNING

Read, annotate and explain

Choose a purpose

Decide whether a particular book may be loaned.

Keep essential facts

ISBN identifies the book, and Available indicates whether it can be borrowed.

Evaluate omissions

The model cannot answer questions about book weight or appearance because those attributes are omitted.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Model a bus network for finding the route with the fewest stops.

Predict first

Model each bus stop as a vertex and each permitted connection as an edge.

Model and check

Vehicle colour is irrelevant. To find fastest rather than fewest-stop routes, add journey-time weights.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Identify two relevant and two irrelevant attributes for a stock checker.

Question 2

Give a simple abstraction for detecting school room timetable clashes: three needed fields and two omitted details.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Why should a model not contain every real-world detail? Give an example where a previously omitted detail becomes relevant.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A good abstraction must record every real-world property.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.